

Lecture(6)

Chapter(2)

Set Theory

- Introduction
- Definition of sets

Introduction

The theory of sets is a language that is perfectly suited to describing and explaining all types of mathematical structures.

Introduction

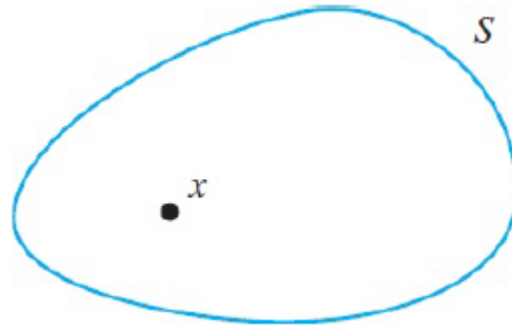
- We will explore different ways that the elements of a set can be related to each other or to the elements of another set.
- These relationships can be described by mathematical objects such as functions, relations, and graphs.
- Our goal is to develop the ability to see mathematical relationships between objects, which in turn will enable us to apply tools from discrete mathematics.

Introduction

- Sets are used to **group objects together**. Objects in a set have similar properties.
- For instance, all the students who are currently enrolled in your class make up a set. Likewise, all the students currently taking a course in discrete mathematics at any class make up a set.
- The language of sets is a means to study such collections in an organized fashion.
- We now provide a definition of a set.

Definition of sets

The simplest way to describe a collection of related objects is as a set.



- Think of the set S as a container where an object x is something that S contains.
- We write $x \in S$ to denote that x is contained in S .
- We also say that " x is a member of S " " x is an element of S " or more simply, " x is in S ."

Definition of sets

Definition:

A *set* is an unordered collection of objects, called *elements* or *members* of the set. A set is said to *contain* its elements. We write $a \in A$ to denote that a is an element of the set A . The notation $a \notin A$ denotes that a is not an element of the set A .

Example:

$\mathbf{N} = \{0, 1, 2, 3, \dots\}$, the set of **natural numbers**

$\mathbf{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$, the set of **integers**

$\mathbf{Z}^+ = \{1, 2, 3, \dots\}$, the set of **positive integers**

$\mathbf{Q} = \{p/q \mid p \in \mathbf{Z}, q \in \mathbf{Z}, \text{ and } q \neq 0\}$, the set of **rational numbers**

$\mathbf{R}$, the set of **real numbers**

$\mathbf{R}^+$, the set of **positive real numbers**

$\mathbf{C}$, the set of **complex numbers**.

It is common for sets to be denoted using uppercase letters. Lowercase letters are usually used to denote elements of sets.

Definition of sets

Membership and Containment

- We can describe examples of sets by listing the elements in the set or by describing the properties that an element in the set has.
- To say that set S consists of the elements $x_1, x_2, \dots, x_n$, we write

$$S = \{x_1, x_2, \dots, x_n\}.$$

- Suppose there is some **property** p that some of the elements of a set S have. We can describe the set of all elements of S that have property p as

$$\{x \in S \mid x \text{ has property } p\}.$$

This is sometimes called "**set builder**" notation, because it explains how to build a list of all the elements of a set.

Definition of sets

Example:

Let $A = \{1, 2, 3, 4, 5, 6, 7, 8\}$. Then $2 \in A$ and $9 \notin A$. If

$$B = \{x \in A \mid x \text{ is odd}\},$$

then the elements of B are

Example: describe a set is to use set builder notation.

- The set of all real numbers can be written as
- The set of all odd positive integers less than 10

Definition of sets

Example: here are some further illustrations of set-builder notation.

1. $\{n : n \text{ is a prime number}\} = \{2, 3, 5, 7, 11, 13, 17, \dots\}$
2. $\{n \in \mathbb{N} : n \text{ is prime}\} = \{2, 3, 5, 7, 11, 13, 17, \dots\}$
3. $\{n^2 : n \in \mathbb{Z}\} = \{0, 1, 4, 9, 16, 25, \dots\}$
4. $\{x \in \mathbb{R} : x^2 - 2 = 0\} = \{\sqrt{2}, -\sqrt{2}\}$
5. $\{x \in \mathbb{Z} : x^2 - 2 = 0\} = \emptyset$
6. $\{x \in \mathbb{Z} : |x| < 4\} = \{-3, -2, -1, 0, 1, 2, 3\}$
7. $\{2x : x \in \mathbb{Z}, |x| < 4\} = \{-6, -4, -2, 0, 2, 4, 6\}$
8. $\{x \in \mathbb{Z} : |2x| < 4\} = \{-1, 0, 1\}$

Definition of sets

Equal sets

Definition:

Two sets are *equal* if and only if they have the same elements. Therefore, if A and B are sets, then A and B are equal if and only if $\forall x(x \in A \leftrightarrow x \in B)$. We write $A = B$ if A and B are equal sets.

Example:

The sets $\{1, 3, 5\}$ and $\{3, 5, 1\}$ are equal, because they have the same elements. Note that the order in which the elements of a set are listed does not matter. Note also that it does not matter if an element of a set is listed more than once, so $\{1, 3, 3, 3, 5, 5, 5, 5\}$ is the same as the set $\{1, 3, 5\}$ because they have the same elements. 

Definition of sets

THE EMPTY SET There is a special set that has no elements. This set is called the **empty set**, or **null set**, and is denoted by $\emptyset$. The empty set can also be denoted by $\{ \}$ (that is, we represent the empty set with a pair of braces that encloses all the elements in this set). Often, a set of elements with certain properties turns out to be the null set. For instance, the set of all positive integers that are greater than their squares is the null set.

Definition of sets

Size of a set

Definition:

Let S be a set. If there are exactly n distinct elements in S where n is a nonnegative integer, we say that S is a *finite set* and that n is the *cardinality* of S . The cardinality of S is denoted by $|S|$.

Example:

Let A be the set of odd positive integers less than 10. Then $|A| = 5$.

Let S be the set of letters in the English alphabet. Then $|S| = 26$.

Because the null set has no elements, it follows that $|\emptyset| = 0$.

A set is said to be *infinite* if it is not finite.

Example:

The set of positive integers is infinite.

Definition of sets

Subsets

Definition:

The set A is a *subset* of B if and only if every element of A is also an element of B . We use the notation $A \subseteq B$ to indicate that A is a subset of the set B .

We see $A \subseteq B$ that if and only if the quantification $\forall x(x \in A \rightarrow x \in B)$

Showing that A is a Subset of B To show that $A \subseteq B$, show that if x belongs to A then x also belongs to B .

Showing that A is Not a Subset of B To show that $A \not\subseteq B$, find a single $x \in A$ such that $x \notin B$.

Fact

The null set ϕ is a subset of every set, that is $\phi \subseteq A$ whenever A is set.

Note that every set is a subset of itself.

Definition of sets

Example: be sure you understand why each of the following is true.

1. $\{2, 3, 7\} \subseteq \{2, 3, 4, 5, 6, 7\}$
2. $\{2, 3, 7\} \not\subseteq \{2, 4, 5, 6, 7\}$
3. $\{2, 3, 7\} \subseteq \{2, 3, 7\}$
4. $\{2n : n \in \mathbb{Z}\} \subseteq \mathbb{Z}$

Fact

If a finite set has n elements, then it has 2^n subsets.

Definition of sets

Power Set

A Power Set is a set of all the subsets of a set and denoted by $\mathcal{P}(S)$.

Example:

For the set $\{a,b,c\}$:

- These are subsets: $\{a\}$, $\{b\}$ and $\{c\}$
- And these are subsets: $\{a,b\}$, $\{a,c\}$ and $\{b,c\}$
- And $\{a,b,c\}$ is also a subset of $\{a,b,c\}$
- And the empty set $\{\}$ is a subset of $\{a,b,c\}$

And when we list all the subsets of $S = \{a,b,c\}$ we get the Power Set of $\{a,b,c\}$:

$$\mathcal{P}(S) = \{ \{\}, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\} \}$$

How Many Subsets

Easy! If the original set has n members, then **the Power Set will have 2^n members.**

Example:

in the $\{a,b,c\}$ example above, there are three members (a,b and c).

So, the Power Set should have $2^3 = 8$, which it does!

Definition of sets

Cartesian Product of Sets

The Cartesian product (or cross product) of A and B , denoted by $A \times B$, is the set

$$A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}$$

Example:

If $A = \{2, 3, 4\}$ and $B = \{4, 5\}$

a) $A \times B = \{(2, 4), (2, 5), (3, 4), (3, 5), (4, 4), (4, 5)\}$

b) $B \times A = \{(4, 2), (4, 3), (4, 4), (5, 2), (5, 3), (5, 4)\}$

Exercises

List the members of these sets.

- a) $\{x \mid x \text{ is a real number such that } x^2 = 1\}$
- b) $\{x \mid x \text{ is a positive integer less than } 12\}$
- c) $\{x \mid x \text{ is the square of an integer and } x < 100\}$
- d) $\{x \mid x \text{ is an integer such that } x^2 = 2\}$

Use set builder notation to give a description of each of these sets,

- a) $\{0, 3, 6, 9, 12\}$
- b) $\{-3, -2, -1, 0, 1, 2, 3\}$

For each of these pairs of sets, determine whether the first is a subset of the second, the second is a subset of the first.

- a) the set of airline flights from New York to New Delhi,
the set of nonstop airline flights from New York to
New Delhi

Exercises

Determine whether each of these pairs of sets are equal.

a) $\{1, 3, 3, 3, 5, 5, 5, 5, 5\}, \{5, 3, 1\}$

b) $\{\{1\}\}, \{1, \{1\}\}$

c) $\emptyset, \{\emptyset\}$

List all the subsets of the following sets.

a. $\{1, 2, 3, 4\}$

b. $\{\emptyset\}$